

四合一无感 FOC 电调 · 双 STM32G431CBT6 + 四 DRV8313

3S / 单板四电机 / CAN 主接口 + 硬件 GLOBAL_ARM / Rev D · 2026-10-09

同名网络为电连接；外围元件表逐项给出两端网络。主控 A→M1/M2，主控 B→M3/M4。不得按旧 F103 图接线。

01 电机 1 / U11 功率级

JM1: 1=U / 2=V / 3=W · HTSSOP28+EP29 · 峰值2.5A

U11 DRV8313PWPR			
M1_CPL	1 CPL	28 GND	GND
M1_CPH	2 CPH	27 IN1	M1_PWM_U
M1_VCP	3 VCP	26 EN1	M1_EN
VM	4 VM	25 IN2	M1_PWM_V
M1_U	5 OUT1	24 EN2	M1_EN
M1_SU	6 PGND1	23 IN3	M1_PWM_W
M1_SV	7 PGND2	22 EN3	M1_EN
M1_V	8 OUT2	21 NC	NC
M1_W	9 OUT3	20 GND	GND
GND	10 PGND3	19 nCOMPO	GND
VM	11 VM	18 nFAULT	M1_FAULT_N
GND	12 COMPP	17 nSLEEP	M1_SLEEP_N
GND	13 COMPN	16 nRESET	M1_RESET_N
GND	14 GND	15 V3P3	M1_DRV3V3
		29 EP	GND

器件电气连接			
VM		100nF/35V	GND
VM		100nF/35V	GND
VM		22uF/35V	GND
VM		10nF/35V	GND
M1_CPH		100nF/35V	M1_CPL
M1_VCP		100nF/35V	VM
M1_DRV3V3		470nF/10V	GND
M1_SU		10mΩ/1%≥0.5W	GND
M1_SV		10mΩ/1%≥0.5W	GND

02 电机 2 / U12 功率级

JM2: 1=U / 2=V / 3=W · HTSSOP28+EP29 · 峰值2.5A

U12 DRV8313PWPR			
M2_CPL	1 CPL	28 GND	GND
M2_CPH	2 CPH	27 IN1	M2_PWM_U
M2_VCP	3 VCP	26 EN1	M2_EN
VM	4 VM	25 IN2	M2_PWM_V
M2_U	5 OUT1	24 EN2	M2_EN
M2_SU	6 PGND1	23 IN3	M2_PWM_W
M2_SV	7 PGND2	22 EN3	M2_EN
M2_V	8 OUT2	21 NC	NC
M2_W	9 OUT3	20 GND	GND
GND	10 PGND3	19 nCOMPO	GND
VM	11 VM	18 nFAULT	M2_FAULT_N
GND	12 COMPP	17 nSLEEP	M2_SLEEP_N
GND	13 COMPN	16 nRESET	M2_RESET_N
GND	14 GND	15 V3P3	M2_DRV3V3
		29 EP	GND

器件电气连接			
VM		100nF/35V	GND
VM		100nF/35V	GND
VM		22uF/35V	GND
VM		10nF/35V	GND
M2_CPH		100nF/35V	M2_CPL
M2_VCP		100nF/35V	VM
M2_DRV3V3		470nF/10V	GND
M2_SU		10mΩ/1%≥0.5W	GND
M2_SV		10mΩ/1%≥0.5W	GND

03 电机 3 / U13 功率级

JM3: 1=U / 2=V / 3=W · HTSSOP28+EP29 · 峰值2.5A

U13 DRV8313PWPR			
M3_CPL	1 CPL	28 GND	GND
M3_CPH	2 CPH	27 IN1	M3_PWM_U
M3_VCP	3 VCP	26 EN1	M3_EN
VM	4 VM	25 IN2	M3_PWM_V
M3_U	5 OUT1	24 EN2	M3_EN
M3_SU	6 PGND1	23 IN3	M3_PWM_W
M3_SV	7 PGND2	22 EN3	M3_EN
M3_V	8 OUT2	21 NC	NC
M3_W	9 OUT3	20 GND	GND
GND	10 PGND3	19 nCOMPO	GND
VM	11 VM	18 nFAULT	M3_FAULT_N
GND	12 COMPP	17 nSLEEP	M3_SLEEP_N
GND	13 COMPN	16 nRESET	M3_RESET_N
GND	14 GND	15 V3P3	M3_DRV3V3
		29 EP	GND

器件电气连接			
VM		100nF/35V	GND
VM		100nF/35V	GND
VM		22uF/35V	GND
VM		10nF/35V	GND
M3_CPH		100nF/35V	M3_CPL
M3_VCP		100nF/35V	VM
M3_DRV3V3		470nF/10V	GND
M3_SU		10mΩ/1%≥0.5W	GND
M3_SV		10mΩ/1%≥0.5W	GND

04 电机 4 / U14 功率级

JM4: 1=U / 2=V / 3=W · HTSSOP28+EP29 · 峰值2.5A

U14 DRV8313PWPR			
M4_CPL	1 CPL	28 GND	GND
M4_CPH	2 CPH	27 IN1	M4_PWM_U
M4_VCP	3 VCP	26 EN1	M4_EN
VM	4 VM	25 IN2	M4_PWM_V
M4_U	5 OUT1	24 EN2	M4_EN
M4_SU	6 PGND1	23 IN3	M4_PWM_W
M4_SV	7 PGND2	22 EN3	M4_EN
M4_V	8 OUT2	21 NC	NC
M4_W	9 OUT3	20 GND	GND
GND	10 PGND3	19 nCOMPO	GND
VM	11 VM	18 nFAULT	M4_FAULT_N
GND	12 COMPP	17 nSLEEP	M4_SLEEP_N
GND	13 COMPN	16 nRESET	M4_RESET_N
GND	14 GND	15 V3P3	M4_DRV3V3
		29 EP	GND

器件电气连接			
VM		100nF/35V	GND
VM		100nF/35V	GND
VM		22uF/35V	GND
VM		10nF/35V	GND
M4_CPH		100nF/35V	M4_CPL
M4_VCP		100nF/35V	VM
M4_DRV3V3		470nF/10V	GND
M4_SU		10mΩ/1%≥0.5W	GND
M4_SV		10mΩ/1%≥0.5W	GND

05A U1 / 双电机控制器 A

STM32G431CBT6 · LQFP48 · 128KB Flash / 32KB SRAM

U1 STM32G431CBT6			
M1_IU	8 PA0 / ADC1.1	30 PA8 / TIM1_CH1	M1_PWM_U
M1_IV	9 PA1 / ADC2.2	31 PA9 / TIM1_CH2	M1_PWM_V
M2_IU	16 PB0 / ADC1.15	32 PA10 / TIM1_CH3	M1_PWM_W
M2_IV	14 PA6 / ADC2.3	39 PA15 / TIM8_CH1	M2_PWM_U
A_VBUS	12 PA4 / ADC2.17	45 PB8 / TIM8_CH2	M2_PWM_V
A_NTC	13 PA5 / ADC2.13	46 PB9 / TIM8_CH3	M2_PWM_W
A_UART_TX	10 PA2 / USART2_TX	26 PB12 / USART2_TX	M1_FAULT_N
A_UART_RX	11 PA3 / USART2_RX	44 PB7 / TIM8_BKIN	M2_FAULT_N
A_CAN_RX	33 PA11 / CAN_RX	22 PB10 / GPIO	M1_RUN
A_CAN_TX	34 PA12 / CAN_TX	25 PB11 / GPIO	M2_RUN
A_SWDIO	37 PA13 / SWDIO	27 PB13 / GPIO	M1_SLEEP_N
A_SWCLK	38 PA14 / SWCLK	28 PB14 / GPIO	M1_RESET_N
A_OSC_IN	5 PF8 / HSE_IN	29 PB15 / GPIO	M2_SLEEP_N
A_OSC_OUT	6 PF1 / HSE_OUT	18 PB2 / GPIO	M2_RESET_N
A_NRST	7 PG10 / NRST	2 PC13 / GPIO	ARM_SAFE
+3V3_A	1 VBAT	15 PA7 / unused	NC
+3V3_B	24,36,48 VDD	17 PB1 / unused	NC
A_VDDA	21 VDDA	3,4 PC14/PC15	NC
A_VDDA	20 VREF+	40,41,42,43 PB3/4/5/6	NC
GND	19,23,35,47 VSS/A		

器件电气连接			
+3V3_A		100nF/10V	GND
+3V3_A		100nF/10V	GND
+3V3_A		100nF/10V	GND
+3V3_A		4.7uF/10V	GND
+3V3_A		100nF/10V	GND
+3V3_A		1uF/10V	A_VDDA
A_VDDA		100nF/10V	GND
A_VDDA			GND

05B U2 / 双电机控制器 B

STM32G431CBT6 · LQFP48 · 128KB Flash / 32KB SRAM

U2 STM32G431CBT6			
M3_IU	8 PA0 / ADC1.1	30 PA8 / TIM1_CH1	M3_PWM_U
M3_IV	9 PA1 / ADC2.2	31 PA9 / TIM1_CH2	M3_PWM_V
M4_IU	16 PB0 / ADC1.15	32 PA10 / TIM1_CH3	M3_PWM_W
M4_IV	14 PA6 / ADC2.3	39 PA15 / TIM8_CH1	M4_PWM_U
B_VBUS	12 PA4 / ADC2.17	45 PB8 / TIM8_CH2	M4_PWM_V
B_NTC	13 PA5 / ADC2.13	46 PB9 / TIM8_CH3	M4_PWM_W
B_UART_TX	10 PA2 / USART2_TX	26 PB12 / TIM1_BKIN	M3_FAULT_N
B_UART_RX	11 PA3 / USART2_RX	44 PB7 / TIM8_BKIN	M4_FAULT_N
B_CAN_RX	33 PA11 / CAN_RX	22 PB10 / GPIO	M3_RUN
B_CAN_TX	34 PA12 / CAN_TX	25 PB11 / GPIO	M4_RUN
B_SWDIO	37 PA13 / SWDIO	27 PB13 / GPIO	M3_SLEEP_N
B_SWCLK	38 PA14 / SWCLK	28 PB14 / GPIO	M3_RESET_N
B_OSC_IN	5 PF8 / HSE_IN	29 PB15 / GPIO	M4_SLEEP_N
B_OSC_OUT	6 PF1 / HSE_OUT	18 PB2 / GPIO	M4_RESET_N
B_NRST	7 PG10 / NRST	2 PC13 / GPIO	B_ARM_FB
+3V3_B	1 VBAT	15 PA7 / unused	NC
+3V3_B	24,36,48 VDD	17 PB1 / unused	NC
B_VDDA	21 VDDA	3,4 PC14/PC15	NC
B_VDDA	20 VREF+	40,41,42,43 PB3/4/5/6	NC
GND	19,23,35,47 VSS/A		

器件电气连接			
+3V3_B		100nF/10V	GND
+3V3_B		100nF/10V	GND
+3V3_B		100nF/10V	GND
+3V3_B		4.7uF/10V	GND
+3V3_B		100nF/10V	GND
+3V3_B		1uF/10V	B_VDDA
B_VDDA		100nF/10V	GND
B_VDDA			GND

06 CAN 总线与外部接口

TCAN1042HGVDR · VCC=5V / VIO=对应 MCU 3.3V

U51 TCAN1042HGVDR			
A_CAN_TX	1 TXD	7 CANH	CANH
A_CAN_RX	4 RXD	6 CANL	CANL
+3V3_A	5 VIO	3 VCC	+5V
GND	2 GND	8 STB	GND

U52 TCAN1042HGVDR			
B_CAN_TX	1 TXD	7 CANH	CANH
B_CAN_RX	4 RXD	6 CANL	CANL
+3V3_B	5 VIO	3 VCC	+5V
GND	2 GND	8 STB	GND

器件电气连接			
A_CAN_TX		10k	
A_CAN_TX		10k	
+3V3_A		100nF/10V	+3V3_A
+5V		100nF/10V	GND
+3V3_A		100nF/10V	GND
+5V		100nF/10V	GND
+3V3_B		1200Ω/25W	GND
CANH		焊接 / 默认闭合	TERM_L
器件电气连接		DNP / 5V输出选择	5V_OUT

J2: 1 CANH / 2 CANL / 3 GND / 4 ARM_EXT / 5 5V_OUT

D3: pin1 CANH / pin2 CANL / pin3 GND，紧贴接口。

120Ω 仅总线两端各一颗；两 MCU TX 不可直接并联。

07 3S 输入 / 5V 降压 / 双路 3.3V

稳态 VM 9~12.6V；VM 尖峰实测 <28V。

J1: BAT+→F1→Q1 D(5~8)；Q1 S(1~3)→VM。

Q1 G(4)=RP_GATE；D2 K=VM / A=RP_GATE。

D1: K=VM / A=GND；C610 为母线储能。

器件电气连接			
BAT+		10A熔断 / 暂定	BAT_FUSED
RP_GATE		100k	GND
VM		470uF/3V 低ESR	GND

U61 TPS54302DDCR			
VM	3 VIN	2 SW	SW5
BUCK_EN	5 EN	6 BOOT	BST5
GND	1 GND	4 FB	FB5

器件电气连接			
VM		10uF/35V	GND
VM		100nF/35V	GND
VM		100nF/16V	GND
BST5		10uH / Lsat≥4A	SW5
SW5		100nF/10V	+5V
+5V		22uF/25V	GND
+5V		22uF/25V	GND
+5V		100k/1%	GND
+5V		13.3k/1%	FB5
FB5		750pF/0G	GND
+5V		511k/1%	FB5
VM		105k/1%	GND
BUCK_EN			BUCK_EN

U62 / U63: TLV75533PDBVR, SOT23-5 (各一路)。			
U62: 1 IN+=+5V / 2 GND / 3 EN+=+5V / 4 NC / 5 OUT+=+3V3_A			
U63: 1 IN+=+5V / 2 GND / 3 EN+=+5V / 4 NC / 5 OUT+=+3V3_B			
+5V		1uF/10V	GND
+3V3_A		4.7uF/10V	GND
+5V		GND	+5V
+3V3_A		GND	+3V3_B

08A 四路采流 / 控制器 A

两路低侧分流 / 电机；仅在有效采样窗口读取。

U21 INA181A1IDBVR			
M1_SU	3 IN+	1 OUT	M1_OU
GND	4 IN-	6 VS	+3V3_A
A_MID	5 REF	2 GND	GND

M1_OU		1000	
M1_IU		220pF/C0G	
+3V3_A		100nF/10V	

M1_IU		GND	M1_OV
		GND	M1_IV
		GND	+3V3_A

U22 INA181A1IDBVR			
M1_SU	3 IN+	1 OUT	M1_OU
GND	4 IN-	6 VS	+3V3_A
A_MID	5 REF	2 GND	GND

M1_OU		1000	
M1_IU		220pF/C0G	
+3V3_A		100nF/10V	

M1_IU		GND	M1_OV
		GND	M1_IV
		GND	+3V3_A

U23 INA181A1IDBVR			
M2_SU	3 IN+	1 OUT	M2_OU
GND	4 IN-	6 VS	+3V3_A
A_MID	5 REF	2 GND	GND

M2_OU		1000	
M2_IU		220pF/C0G	
+3V3_A		100nF/10V	

M2_IU		GND	M2_OV
		GND	M2_IV
		GND	+3V3_A

U24 INA181A1IDBVR			
M2_SU	3 IN+	1 OUT	M2_OU
GND	4 IN-	6 VS	+3V3_A
A_MID	5 REF	2 GND	GND

M2_OU		1000	
M2_IU		220pF/C0G	
+3V3_A		100nF/10V	

M2_IU		GND	M2_OV
		GND	M2_IV
		GND	+3V3_A

VOUT = MID + 0.2 V/A × I低侧；IN- 对对应 RS 地端独立取样。

08B 四路采流 / 控制器 B

两路低侧分流 / 电机；仅在有效采样窗口读取。

U25 INA181A1IDBVR			
M3_SU	3 IN+	1 OUT	M3_OU
GND	4 IN-	6 VS	+3V3_B
B_MID	5 REF	2 GND	GND

M3_OU		1000	
M3_IU		220pF/C0G	
+3V3_B		100nF/10V	

M3_IU		GND	M3_OV
		GND	M3_IV
		GND	+3V3_B

U26 INA181A1IDBVR			
M3_SU	3 IN+	1 OUT	M3_OU
GND	4 IN-	6 VS	+3V3_B
B_MID	5 REF	2 GND	GND

M3_OU		1000	
M3_IU		220pF/C0G	
+3V3_B		100nF/10V	

M3_IU		GND	M3_OV
		GND	M3_IV
		GND	+3V3_B

U27 INA181A1IDBVR			
M4_SU	3 IN+	1 OUT	M4_OU
GND	4 IN-	6 VS	+3V3_B
B_MID	5 REF	2 GND	GND

M4_OU		1000	
M4_IU		220pF/C0G	
+3V3_B		100nF/10V	

M4_IU		GND	M4_OV
		GND	M4_IV
		GND	+3V3_B

U28 INA181A1IDBVR			
M4_SU	3 IN+	1 OUT	M4_OU
GND	4 IN-	6 VS	+3V3_B
B_MID	5 REF	2 GND	GND

M4_OU		1000	
M4_IU		220pF/C0G	
+3V3_B		100nF/10V	

M4_IU		GND	M4_OV
		GND	M4_IV
		GND	+3V3_B

VOUT = MID + 0.2 V/A × |低侧; IN- 对对应 RS 地端独立取样。